

Problem 1. By Fokker-Planck equation:

$$\frac{\partial}{\partial t} p(x,t) = -\frac{\partial}{\partial x} [f(x,t) p(x,t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} (g^2(x,t) p(x,t))$$

To find $v(x,t)$ s.t. $dx_t = v(x_t, t) dt$, and

$$v(x,t) \text{ satisfies } \frac{\partial}{\partial t} p(x,t) + \frac{\partial}{\partial x} [v(x,t) p(x,t)] = 0$$

(came from conservation of probability)

$$\Rightarrow -\frac{\partial}{\partial x} (f p) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (g^2 p) = -\frac{\partial}{\partial x} (v p)$$

$$\Rightarrow v = f - \frac{1}{2p} \frac{\partial}{\partial x} (g^2 p)$$

$$= f - \frac{1}{2p} \left[\frac{\partial}{\partial x} (g^2) p + g^2 \frac{\partial}{\partial x} p \right]$$

$$= f - \frac{1}{2} \frac{\partial}{\partial x} g^2 - \frac{1}{2} g^2 \frac{\partial}{\partial x} \log p$$

$$\Rightarrow dx_t = \left[f(x_t, t) - \frac{1}{2} \frac{\partial}{\partial x} g^2(x_t, t) - \frac{1}{2} g^2(x_t, t) \frac{\partial}{\partial x} (\log p(x_t, t)) \right] dt$$